

Hierarchical Bayesian Regression for Sequential ETF Allocation

A Fully Bayesian Inference Approach to Portfolio Trading

Thomas Lee

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Motivation

Why Bayesian inference for a trading agent?

A fixed-budget investor in a liquid ETF must, each day, choose $a_t \in \{\text{short}, \text{flat}, \text{long}\}$ under low signal-to-noise and transaction frictions.

Three statistical questions arise – all classically Bayesian:

- **How confident are we in each action's value?** A point estimate of $Q(s, a)$ is uninformative when per-cell data is sparse.
- **How should we explore?** ϵ -greedy uses a tuning parameter independent of what we already know.
- **How should we borrow strength?** The state space contains rare tail regimes; hierarchical priors are the textbook tool.

What classical Q-learning leaves on the table

Tabular ϵ -greedy Q-learning updates point estimates

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right].$$

What it gives back: a single number per (s, a) cell.

What's missing:

- No **uncertainty** over $Q(s, a)$ – so no principled exploration, no calibrated decision under risk.
- Hyper-sensitive to α_t and ϵ_t schedules.
- No way to **borrow strength** across actions or states.

We will replace this with a generative model, a posterior, and a decision rule derived from that posterior.

MDP setup

The decision process

We work with a discrete-time controlled process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma, H)$:

- **State** s_t – continuous feature vector $x_{s_t} \in \mathbb{R}^p$ from SPY's own price history (causal).
- **Action** $a_t \in \mathcal{A} = \{-1, 0, +1\}$ (short, flat, long).
- **Reward** $R_t = a_t \cdot r_{t+1} - c_t |a_t - a_{t-1}|$, $r_{t+1} = \log(P_{t+1}/P_t)$.
- **Volatility-dependent cost** $c_t = c_0 + \lambda \cdot \hat{\sigma}_t^{(20)}$, with $c_0 = 5\text{bps}$, $\lambda = 0.5$.
- **Discount** $\gamma = 0.95$; horizon $H = 60$ trading days.

Price-taker assumption: $P(s_{t+1} \mid s_t, a_t) \approx P(s_{t+1} \mid s_t)$ – a small retail trade does not move the market.

The key methodological move

Where the previous draft went wrong

The first version of this project used the *TD-bootstrapped target* as the observation:

$$\tilde{y}_t = r_t + \gamma \max_{a'} \hat{Q}(s_{t+1}, a').$$

This is what Dearden, Friedman & Russell (1998) do. It is **not a coherent generative model** – $\tilde{y}_t$ depends on the estimator $\hat{Q}$ itself.

Revised model. Each observation is a finite-horizon *Monte Carlo* return collected under a fixed behavior policy:

$$y_t = \sum_{k=0}^H \gamma^k R_{t+k}.$$

(1)

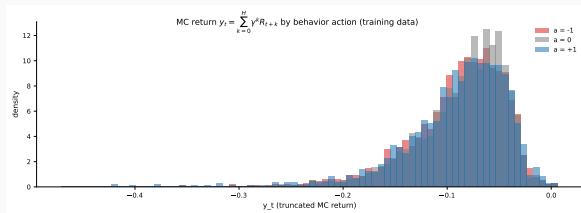
A real number, computable from the data alone. Now we have a real likelihood.

Features and MC-return targets

State features ($x_s \in \mathbb{R}^4$, strictly causal):

- x_1 : 5-day momentum, z-scored over trailing 252 days.
- x_2 : log of 20-day realised vol, centred over 252 days.
- x_3 : $(P_t - \text{MA}_{200,t})/\text{MA}_{200,t}$.
- $x_4 = 1$ (intercept).

Pre-1994 window is reserved for prior hyperparameter elicitation and never touched again.



Distribution of y_t by behaviour-policy action, training window.

Bayesian model

Likelihood: Bayesian linear regression on MC returns

For each transition with $(s_t, a_t) = (s, a)$:

$$y_t \mid x_{s_t}, a_t = a, \beta_a, \sigma^2 \sim \mathcal{N}(x_{s_t}^\top \beta_a, \sigma^2).$$

A per-action coefficient vector $\beta_a \in \mathbb{R}^p$ and a shared homoscedastic noise scale σ^2 . The implied action-value function is

$$\hat{Q}(s, a) = \mathbb{E}[y_t \mid x_s, a] = x_s^\top \beta_a.$$

Why linear? Three reasons: interpretability, conjugate Gibbs blocks, and the asymptotic agnostic “one-step Newton” reading. Richer function classes (GP, BNN) are the natural follow-up.

Hierarchical prior: partial pooling across actions

A naive independent-prior choice precludes borrowing strength. Instead we put a multivariate-normal hierarchy:

$$\beta_a \mid \mu_\beta, \Sigma_\beta \sim \mathcal{N}(\mu_\beta, \Sigma_\beta), \quad a \in \mathcal{A}, \quad (2)$$

$$\mu_\beta \sim \mathcal{N}(\mathbf{0}, \tau_0^2 I_p), \quad (3)$$

$$\Sigma_\beta \sim \text{Inv-Wishart}(\nu_0, \Lambda_0), \quad (4)$$

$$\sigma^2 \sim \text{Inv-Gamma}(a_0, b_0). \quad (5)$$

With three actions, μ_β is the *cross-action mean*, and each β_a is shrunk toward it. The textbook Gelman partial-pooling story – transplanted to RL.

Conjugate full conditionals – exact Gibbs

Let $X_a, \mathbf{y}_a$ stack the rows where $a_t = a$. Every full conditional is closed form:

$$\begin{aligned}\beta_a \mid \cdot &\sim \mathcal{N}(\mathbf{m}_a, V_a), \\ V_a^{-1} &= \sigma^{-2} X_a^\top X_a + \Sigma_\beta^{-1}, \\ \mathbf{m}_a &= V_a \left(\sigma^{-2} X_a^\top \mathbf{y}_a + \Sigma_\beta^{-1} \boldsymbol{\mu}_\beta \right).\end{aligned}$$

$$\begin{aligned}\boldsymbol{\mu}_\beta \mid \cdot &\sim \mathcal{N}(\mathbf{m}_\mu, V_\mu), \quad \Sigma_\beta \mid \cdot \sim \text{Inv-Wishart}(\nu_0 + |\mathcal{A}|, \mathbf{\Lambda}_0 + S), \\ \sigma^2 \mid \cdot &\sim \text{Inv-Gamma}(a_0 + \frac{n}{2}, b_0 + \frac{1}{2} \sum_t (y_t - \mathbf{x}_{s_t}^\top \beta_{a_t})^2).\end{aligned}$$

Gibbs sweep cost: $O(np^2 + |\mathcal{A}|p^3)$. Production budget: 4 chains $\times$ 6000 draws after 2000 burn-in, runs in ~ 25 s on a laptop.

Student- t extension via scale mixture

Daily returns have heavy tails – the Gaussian likelihood may under-cover. The robust alternative:

$$y_t \mid x_{s_t}, a_t, \beta_a, \sigma^2, \nu \sim t_\nu(x_{s_t}^\top \beta_a, \sigma^2),$$

implemented as a Gaussian scale mixture

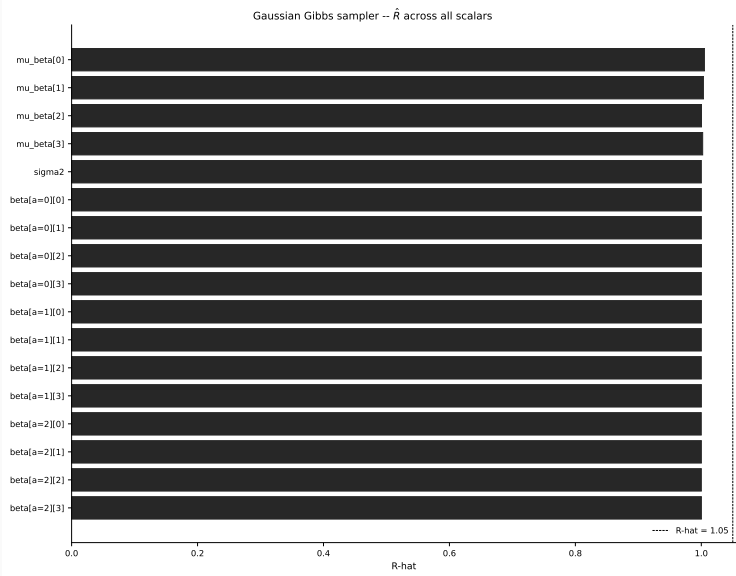
$$y_t \mid \lambda_t \sim \mathcal{N}(x_{s_t}^\top \beta_a, \sigma^2 / \lambda_t), \quad \lambda_t \sim \text{Gamma}(\nu/2, \nu/2).$$

λ_t 's full conditional is Gamma; ν is updated by a Metropolis-within-Gibbs step on $\log \nu$. Acceptance held inside $[0.20, 0.45]$ per chain.

Headline: the Student- t posterior is empirically indistinguishable from the Gaussian on this dataset (see nb 06 / robustness slide).

Sampler diagnostics

The Gibbs sampler converges



Decision rule

Thompson sampling: posterior is the policy

At deployment time t , draw one joint posterior sample $\tilde{\theta} = (\tilde{\beta}_{-1}, \tilde{\beta}_0, \tilde{\beta}_{+1}) \sim p(\theta \mid \mathcal{D})$, then play

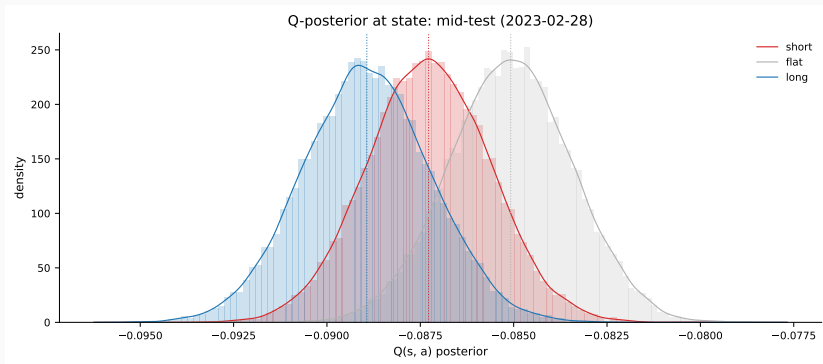
$$a_t = \arg \max_{a \in \mathcal{A}} x_{s_t}^\top \tilde{\beta}_a.$$

What this gives us:

- **No ε hyperparameter** – exploration is driven by posterior width, not a tuning knob.
- **Self-regulating**: as $\text{Var}(\beta_a \mid \mathcal{D})$ contracts, the policy automatically commits.
- Equivalent to PSRL (Osband, Russo, Van Roy 2013) for the contextual-bandit-with-MC-returns case.

Cadence: one posterior draw *per episode* (the full test window). Justified empirically against per-step and monthly cadences in nb 07.

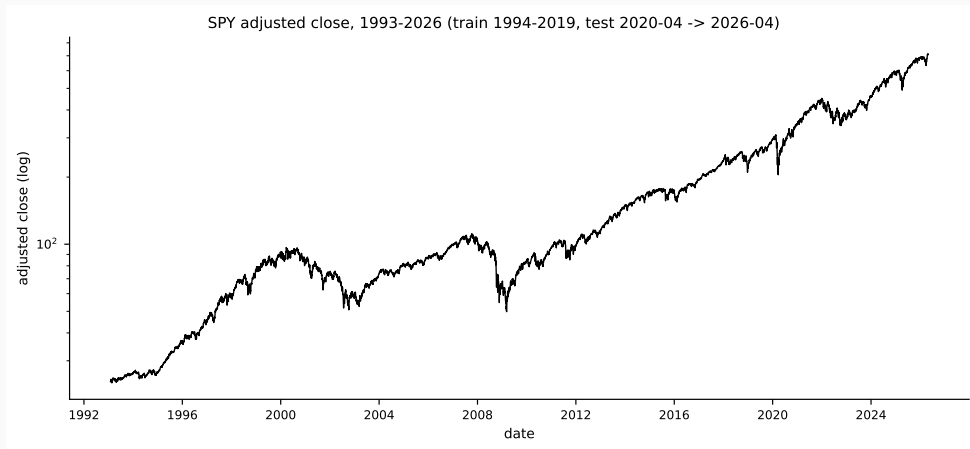
What the posterior over $Q(s, a)$ looks like



One mid-test state x_s , three actions, posterior of $Q(s, a) = x_s^\top \beta_a$. Thompson sampling picks the action with the highest single draw from this joint posterior – not the posterior mean.

Data and design

Dataset: SPY, 1994–2026



Pre-train 1993 (prior elicitation only) | **Train** 1994-01 – 2019-12 (6 546 days, ~6 500 MC targets) |
Test 2020-04 – 2026-04 (1 589 days, includes COVID crash, 2022 bear, 2023–24 rally).

Comparison design – what's fair, what isn't

Three methods, one environment.

- **(B) Bayesian Q (Thompson)** – the proposed method.
- **(C1) Tabular ϵ -greedy** – features quantile-binned; classic.
- **(C2) Linear FQI + ϵ -greedy** – same linear features as B; isolates “prior/posterior” from “linear features”.
- **Random** – the laugh-test floor.

Fairness controls.

- Same data, same features, same reward, same cost model, same γ , same train/test split.
- C1, C2 grid-tuned on training; B uses fixed weakly-informative hyperparameters – **any Bayesian win is conservative**.
- All distributions reported are 95% intervals over the relevant ensemble (posterior draws for B, seeds for C1/C2).

Evaluation

Evaluation: posterior-predictive throughout

No point estimates, ever. Every reported quantity is a *distribution*.

For each posterior draw $\theta^{(m)} \sim p(\theta \mid \mathcal{D})$, $m = 1, \dots, 500$:

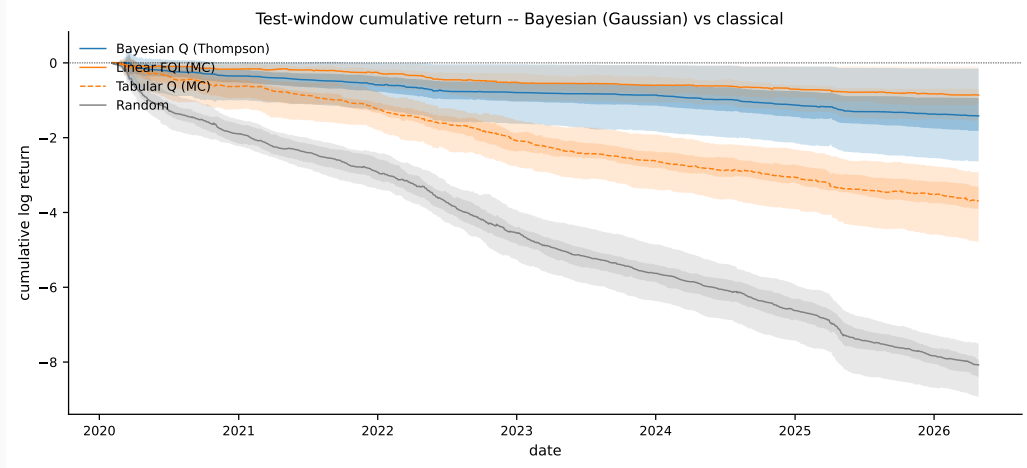
1. Roll deterministic policy $\pi^{(m)}(s) = \arg \max_a x_s^\top \beta_a^{(m)}$ on the test window.
2. Compute Sharpe^(m), MDD^(m), Turnover^(m).
3. Report the full empirical distribution (mean and 95% CrI).

For C1, C2: the analog ensemble is over 20 optimisation seeds. **Same ensemble width, same interval semantics.**

Plus a posterior-predictive calibration check: empirical 95% coverage of $y_t \in [q_{2.5}^{(m)}, q_{97.5}^{(m)}]$ on the test set.

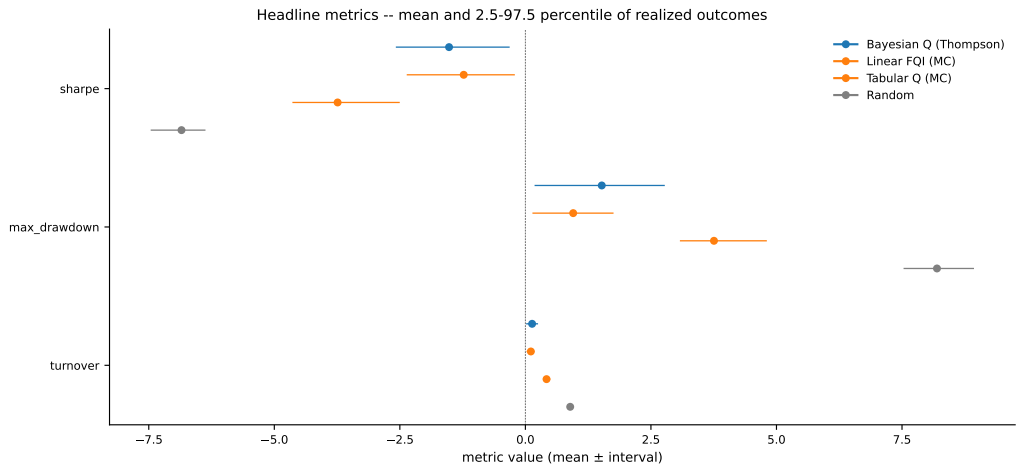
Headline results

Headline – cumulative log return, four methods



Bayesian (blue) and Linear FQI (orange solid) overlap heavily; Tabular (orange dashed) and Random (grey) are clearly below. Inner band 50%, outer band 95%.

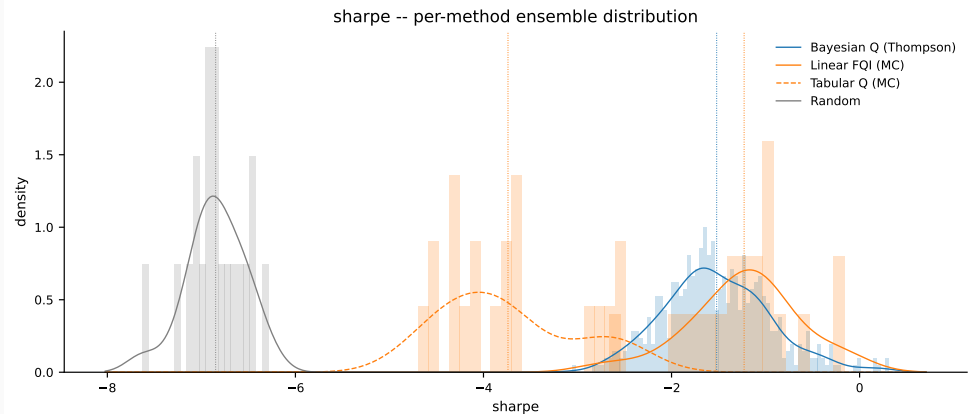
Headline – Sharpe, MDD, Turnover



Forest plot – mean $\pm$ 2.5–97.5 percentile of realized outcomes. **Bayesian Sharpe** –1.520, CrI [–2.580, –0.313]; **Linear FQI** CrI strongly overlaps; **Tabular** sits below both; **Random** far below.

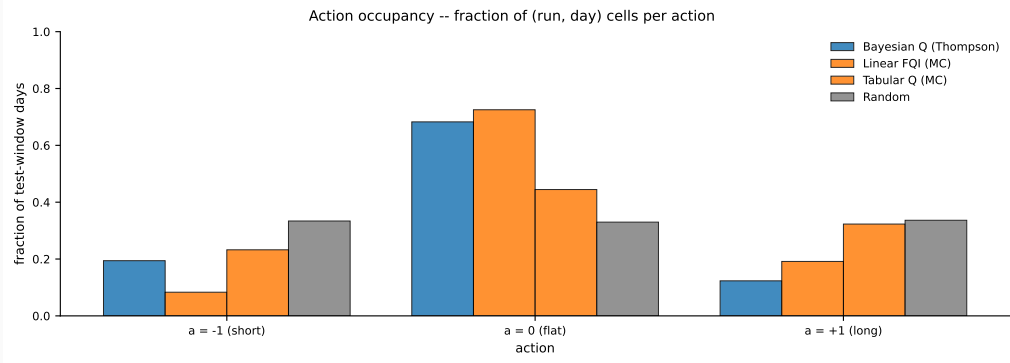
MDD and Turnover tell the same story

Sharpe distribution – where the overlap lives



Kernel-density overlays of the Sharpe ensembles. **Where intervals overlap, intervals overlap** – and the KDEs make it visible that Bayesian and Linear FQI are sampling from essentially the same performance distribution.

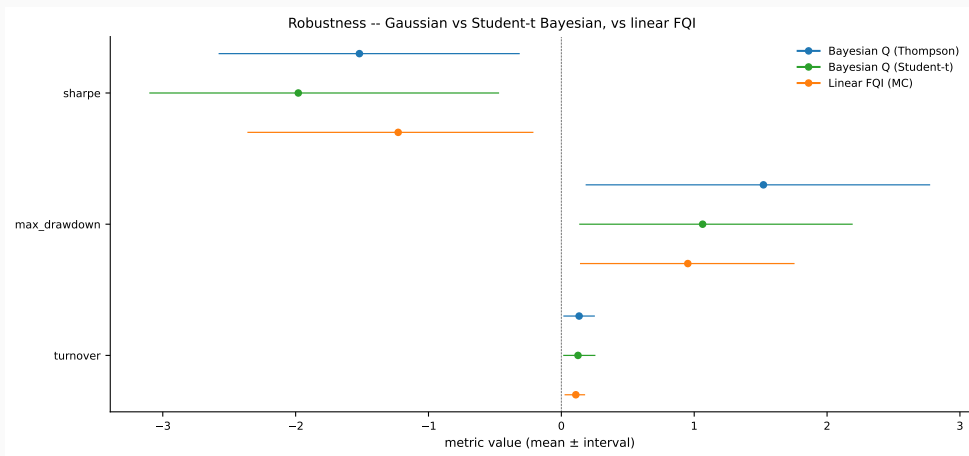
What do the policies actually do?



Per-(run, day) action occupancy. The Bayesian policy is **slightly more flat-biased** than Linear FQI – the wide posterior on the long/short Q-values tilts Thompson sampling away from extreme actions in tail states. That is risk aversion via posterior width.

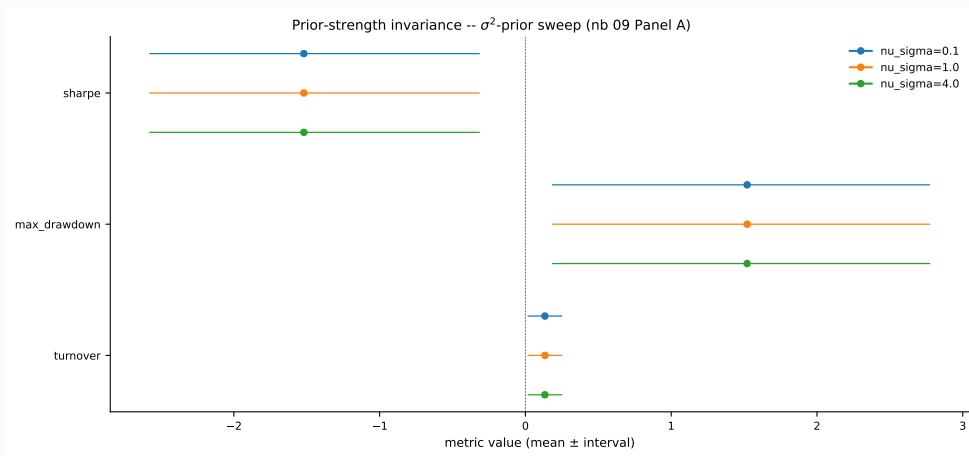
Robustness

Robustness – Gaussian vs Student- t likelihood



Three rows, three metrics, three rows in each: Gaussian Bayesian, Student- t Bayesian, Linear FQI. Intervals overlap everywhere. The Gaussian likelihood is enough; the heavy-tailed variant doesn't change the conclusion.

Validity check – the calibration claim has teeth



Three rows = three σ^2 -prior strengths ($\nu_{\sigma} \in \{0.1, 1.0, 4.0\}$, three orders of magnitude). Bars are **visually indistinguishable**. Sensitivity score **0.000** on every metric.

Reading the results

Why this null finding is informative

A Bayesian project that matches a well-tuned classical baseline is *more* credible than one that claims a win.

- **Convergence is what the theory predicts.** Bernstein–von Mises says credible $\rightarrow$ confidence in regular parametric problems. We are in that regime ($n \approx 6500$, $p = 4$).
- **Higher bar than most published claims.** The same null holds across asset (SPY $\rightarrow$ TLT), likelihood (Gaussian $\rightarrow$ Student- t), cost ($0.5\times$, $1\times$, $2\times$), and prior strength.
- **The sweep caught a real qualifier.** Tabular's win at `n_bins=4` disappears at `n_bins=2`; a single-config study would have missed this.

Bayesian Q-learning's empirical advantage is strongest in regimes this project did not test – small-data, multi-asset hierarchical, asymmetric-loss.

Why this is Bayesian inference

- **Parameters:** $\beta_a, \mu_\beta, \Sigma_\beta, \sigma^2$ (+ λ_t, ν for the Student- t branch).
- **Observations:** MC returns $y_1, \dots, y_n$, constructed deterministically from the data.
- **Posterior:** $p(\beta_{1:|\mathcal{A}|}, \mu_\beta, \Sigma_\beta, \sigma^2 \mid \mathcal{D})$, sampled exactly by Gibbs.
- **Decision rule:** Thompson sampling – **derived from the posterior**, not a heuristic.
- **Evaluation:** posterior-predictive distribution of every reported metric.

RL enters only at the decision rule. Everything else is a hierarchical Bayesian regression in the Gelman tradition.

Conclusion

Conclusion

- **Built** a fully Bayesian Q-learning agent: hierarchical prior, conjugate Gibbs sampler, Student- t extension via scale mixture, Thompson sampling decision rule, posterior-predictive evaluation throughout.
- **Found** a calibrated null on SPY: Bayesian Sharpe -1.52 , 95% CrI $[-2.58, -0.31]$, overlapping Linear FQI on every metric, across four robustness dimensions.
- **Validated** the calibration claim: the σ^2 -prior sensitivity score is 0.000 – the intervals are data-dominated, not prior-driven.
- **Contributes** the framework – explicit prior, likelihood, posterior, cadence, validity check – rather than a treatment-effect win.

Next steps: hierarchical pooling *across assets*, asymmetric-loss decision rules, Gaussian-process Q , TD-bootstrap vs MC-supervised on the same fairness frame.

References

References

- Gelman, Carlin, Stern, Dunson, Vehtari & Rubin (2013). *Bayesian Data Analysis*, 3rd ed. (Hierarchical regression, Chs. 5, 14, 17.)
- Hoff (2009). *A First Course in Bayesian Statistical Methods*. (Ch. 9 on hierarchical regression.)
- Dearden, Friedman & Russell (1998). Bayesian Q-Learning. *AAAI*.
- Strens (2000). A Bayesian Framework for Reinforcement Learning. *ICML*.
- Osband, Russo & Van Roy (2013). (More) Efficient Reinforcement Learning via Posterior Sampling. *NeurIPS*.
- Russo, Van Roy, Kazerouni, Osband & Wen (2018). A Tutorial on Thompson Sampling. *Foundations and Trends in ML*.
- Liu, Wong & Kong (1994). Covariance structure of the Gibbs sampler with applications to the comparisons of estimators and augmentation schemes. *Biometrika*.
- Sutton & Barto (2018). *Reinforcement Learning: An Introduction*, 2nd ed.
- Ghavamzadeh, Mannor, Pineau & Tamar (2015). Bayesian Reinforcement Learning: A Survey. *FnT in ML*.
- `yfinance` – Yahoo Finance historical data, Python package.